Lab 7: Machine Learning 1

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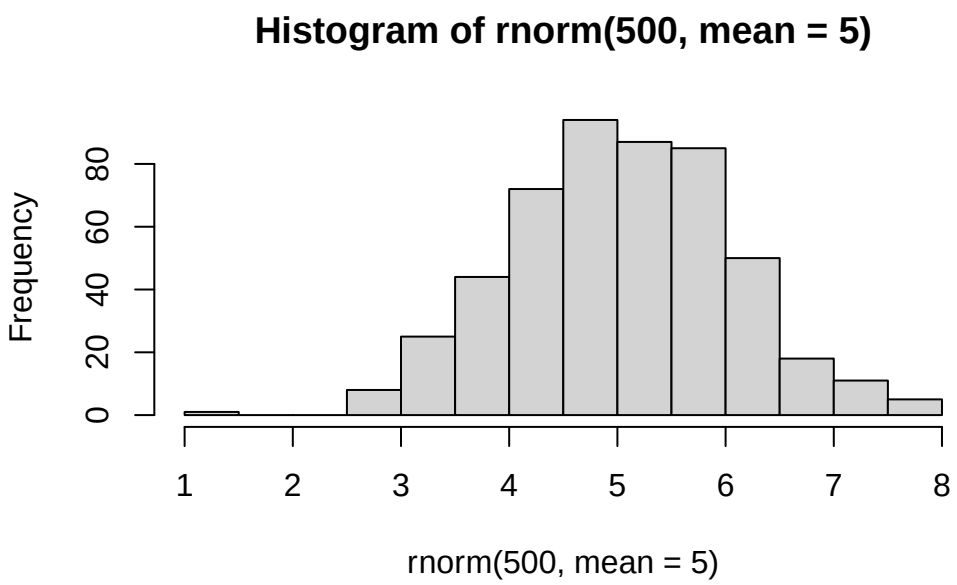
Fundamentals of machine learning including clustering and dimensionality reduction

Creating a dataset:

```
hist(rnorm(500,mean=5))
```




```
x <- c(rnorm(30,mean=-3),rnorm(30,mean=3))
y <- rev(x)

data <- cbind(x,y)
```

K-means clustering

```r
k <- kmeans(data, 2)
k
```

```
K-means clustering with 2 clusters of sizes 30, 30

Cluster means:
          x         y
1 -2.639987  3.041102
2  3.041102 -2.639987

Clustering vector:
 [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2
[39] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2

Within cluster sum of squares by cluster:
[1] 56.81872 56.81872
 (between_SS / total_SS =  89.5 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
```

Question: How many points are in each cluster?

```r
k$size
```

```
[1] 30 30
```

What components of your result object details: - cluster assignment/membership? - cluster center?

```r
k$cluster
```

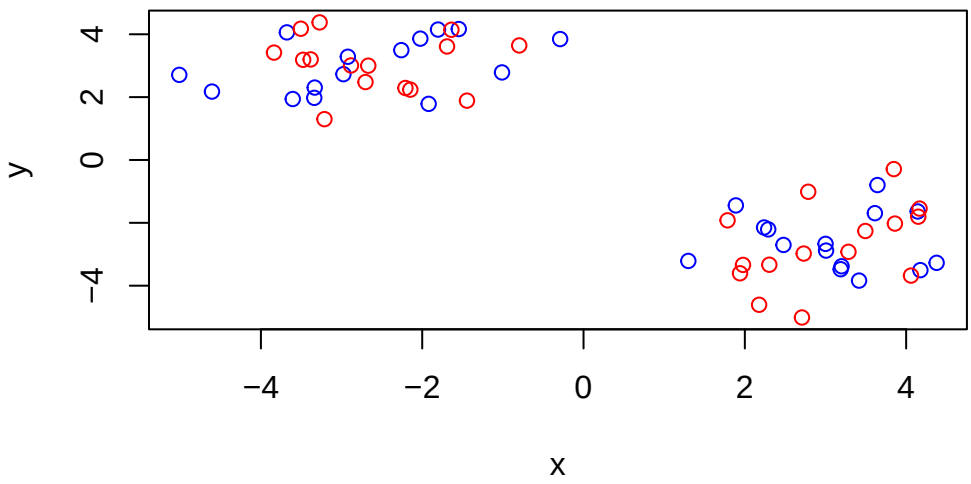
```
 [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2
[39] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
```

```r
k$centers
```

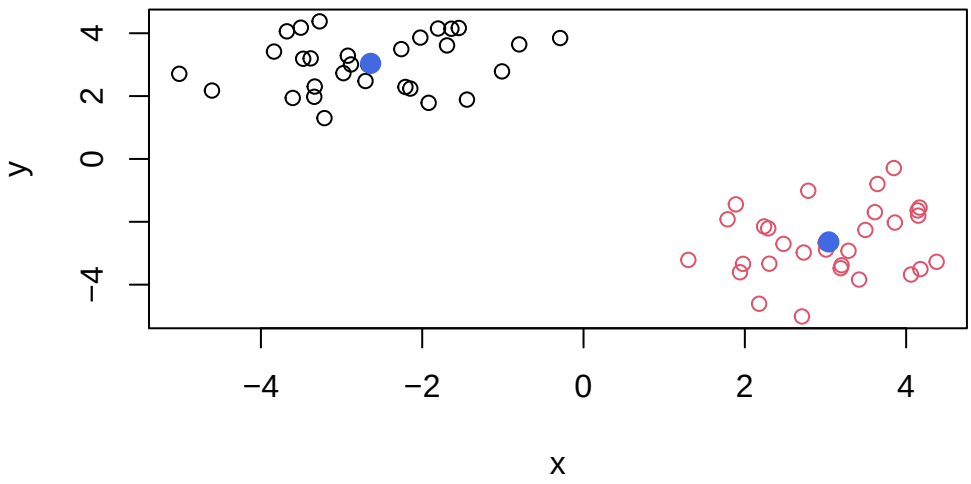
```
          x         y
1 -2.639987  3.041102
2  3.041102 -2.639987
```

Make a clustering results figure of the data colored by cluster membership

```
plot(data, col=c("red", "blue"))
```




```
plot(data, col=k$cluster)
points(k$centers, col="royalblue", pch=20, cex=2)
```



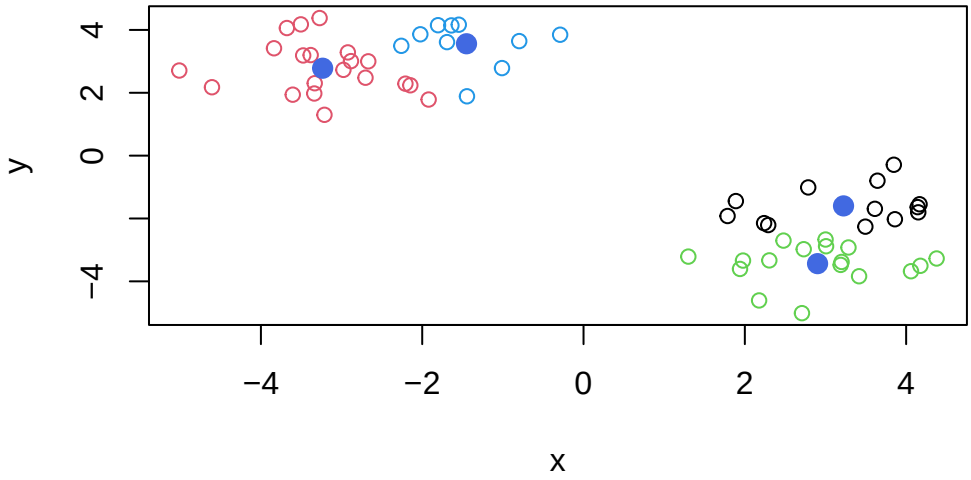

```
# pch is plotting character i.e. dot type
# cex is character expansion
```

K-means clustering is very popular as it is very fast and relatively straightforward: it takes numeric data as input and return the cluster membership, etc.

The problem is that we tell k-means how many clusters we want

Run kmeans again and cluster it into 4 groups and plot the results like above

```
k4 <- kmeans(data, 4)
plot(data, col=k4$cluster)
points(k4$centers, col="royalblue", pch=20, cex=2)
```

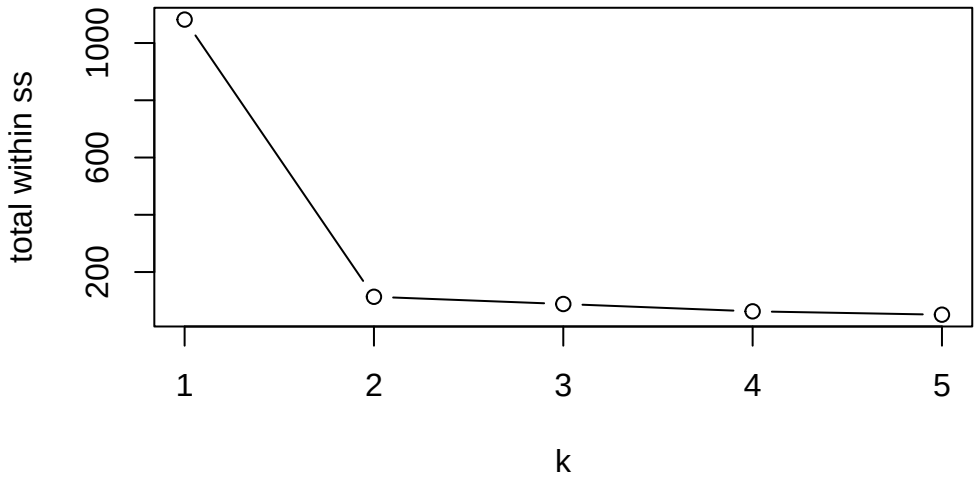


It picks lowest value of total variation within clusters. That is the tightest fit and returns that. There is a large reduction in SS at k=2 but after that the values do not go down as quickly.

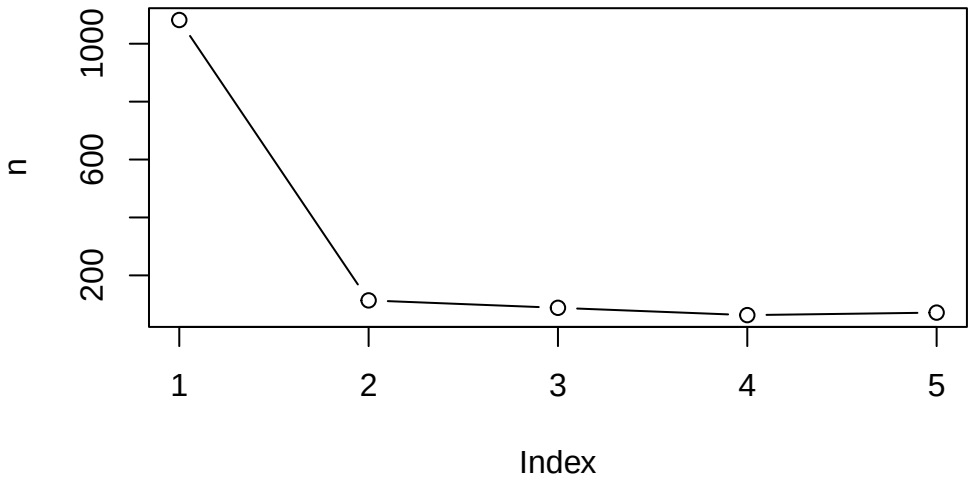
```
k1 <- kmeans(data, 1)
k3 <- kmeans(data, 3)
k5 <- kmeans(data, 5)

ss <- c(k1$tot.withinss, k$tot.withinss, k3$tot.withinss, k4$tot.withinss, k5$tot.withinss)
ss
```

```
[1] 1081.88073  113.63744   88.39794   62.76365   51.06197
```

```
plot(ss, type="b", xlab="k", ylab="total within ss")
```




```
n <- NULL

for (i in 1:5)
{
  n <- c(n, kmeans(data, i)$tot.withinss)
}

plot(n, type="b")
```



Hierarchichal Clustering

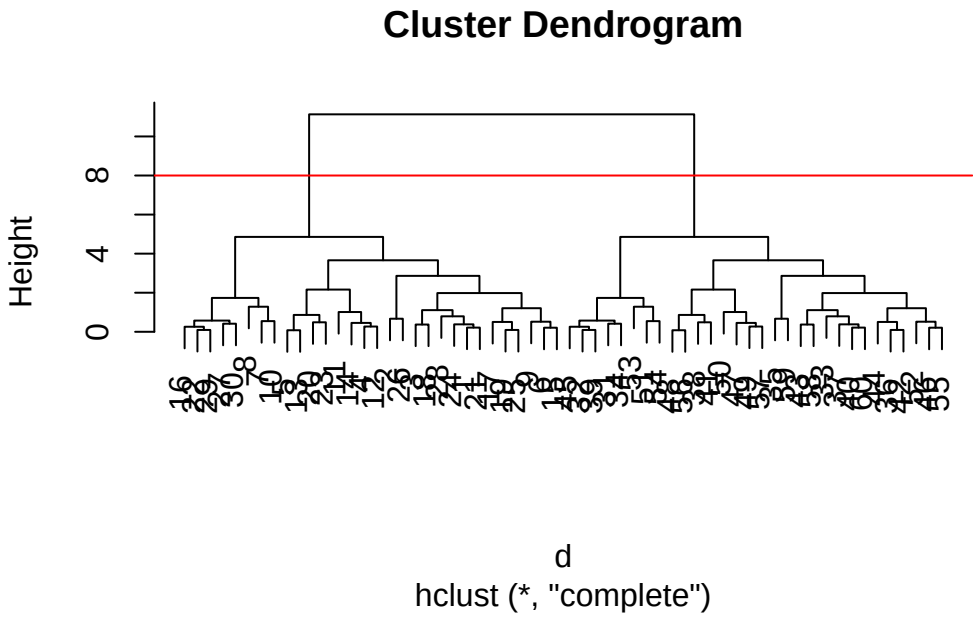
We can't just input our data. We need to first calculate a distance matrix for our data (e.g. `dist()`) and use it as input for the `hclust()` function. Not just euclidean distance but also sequence similarity, amino acid similarity, etc.

```r
d <- dist(data)
# 60x60 matrix with the distance of every point in the dataset with every other point

hc <- hclust(d)
hc
```

```
Call:
hclust(d = d)

Cluster method   : complete
Distance         : euclidean
Number of objects: 60
```

There is a plot method for hclust results

```r
plot(hc)
abline(h=8, col="red")
```



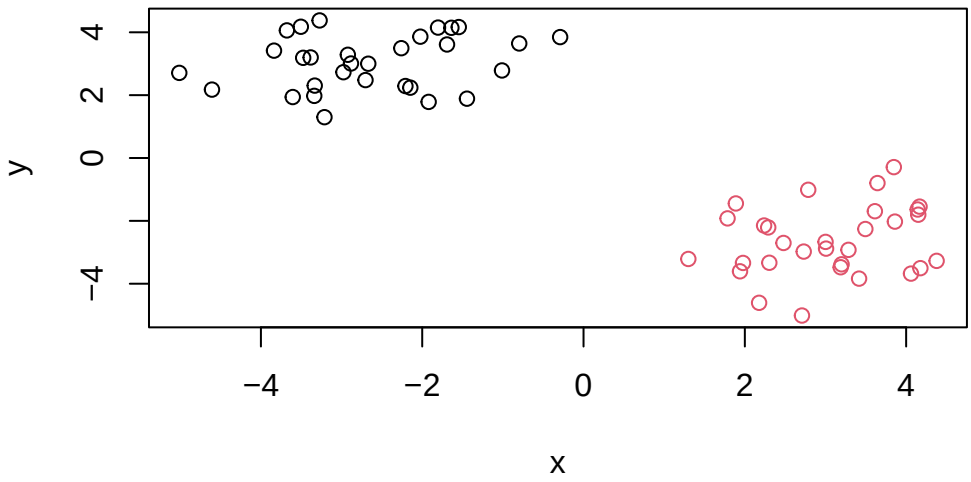
The numbers between 1-30 and 31-60 are on separate sides of the graphs i.e. they are highly separated by distance.

To get our cluster membership vector (i.e. our main clustering result) we can "cut" the tree at a given height or a height that yields a given "k" number of groups.

```
hccol <- cutree(hc, h=8)
```

Plot the data with our hclust result coloring

```
plot(data, col=hccol)
```



Principal Component Analysis

PCA of UK food data

```
url <- "https://tinyurl.com/UK-foods"
food <- read.csv(url, row.names=1)
dim(food)
```

```
[1] 17  4
```

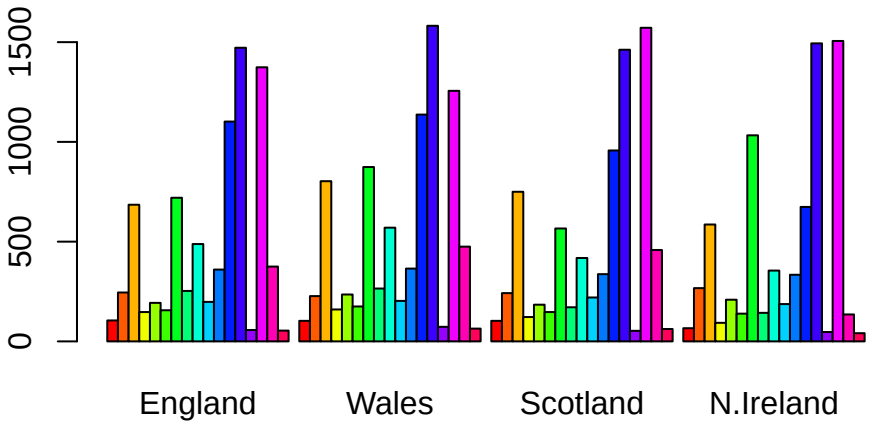
```
head(food)
```

```
              England Wales Scotland N.Ireland
Cheese            105   103      103        66
Carcass_meat      245   227      242       267
Other_meat        685   803      750       586
Fish              147   160      122        93
Fats_and_oils     193   235      184       209
Sugars            156   175      147       139
```

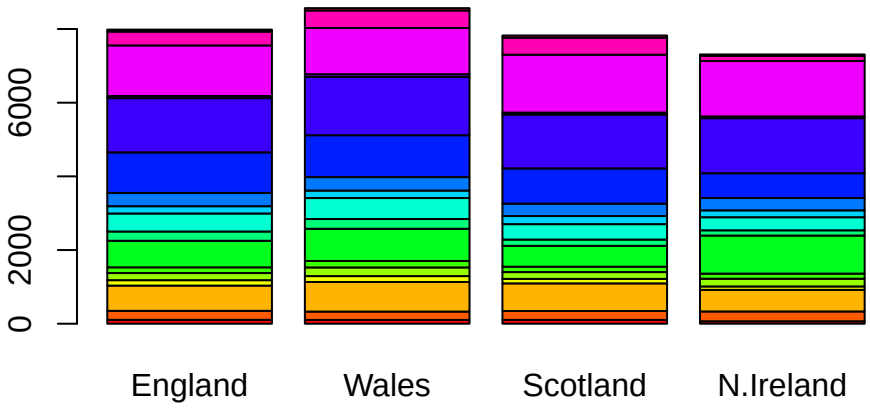
```
# rownames(food) <- food [,1]
# food <- food[,-1]
# food

# Can get disruptive if run again (keeps removing columns)
```

Some base figures

```
barplot(as.matrix(food), beside=T, col=rainbow(nrow(food)))
```

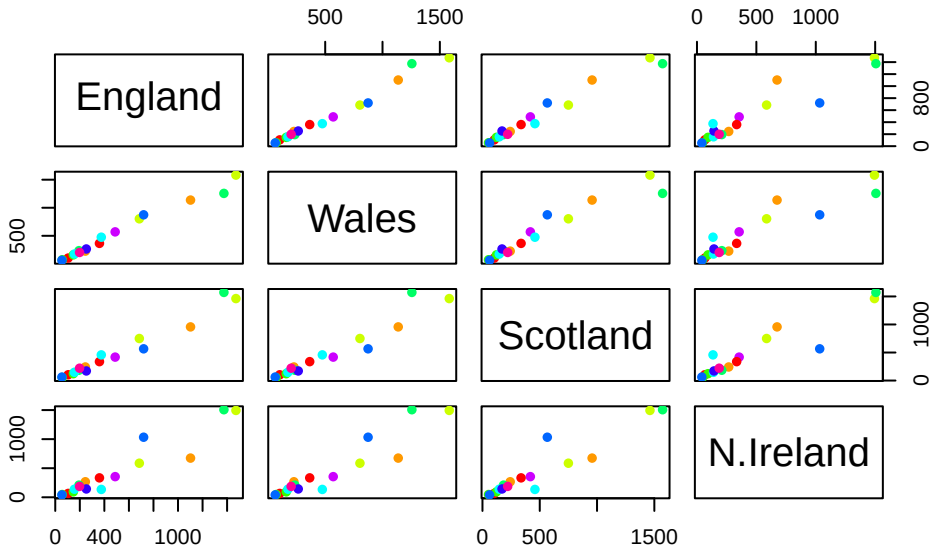



```
barplot(as.matrix(food), beside=F, col=rainbow(nrow(food)))
```



There is one plot that can be useful for small datasets

```
pairs(food, col=rainbow(10), pch=16)
```



It can be challengin to spot major trends and patters even in relatively small multivariate datasets (here we have 17, typically we have 1000s)

PCA

In base R, the main PCA function is `prcomp()`

I will take the transpose of our food data so the "foods" are in the columns

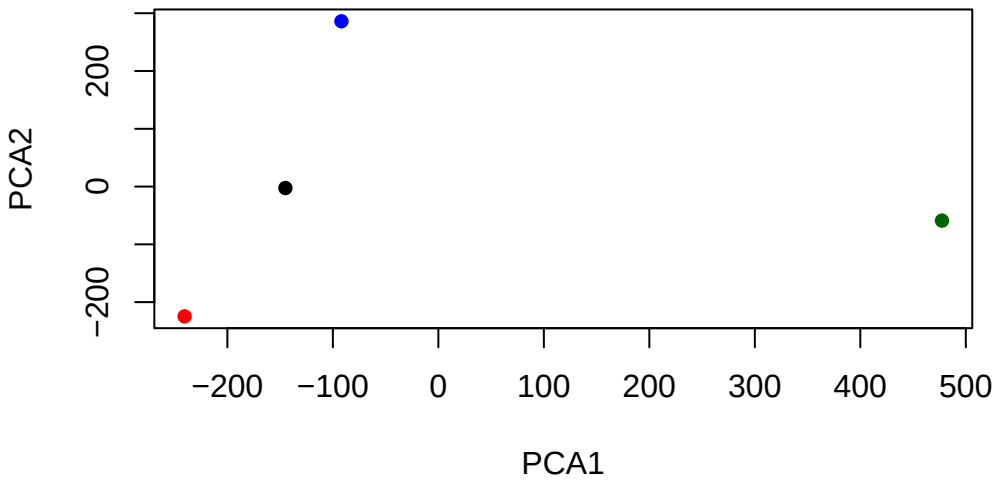
```
pca <- prcomp(t(food))
summary(pca)
```

```
Importance of components:
                            PC1      PC2      PC3       PC4
Standard deviation     324.1502 212.7478 73.87622 2.921e-14
Proportion of Variance   0.6744   0.2905  0.03503 0.000e+00
Cumulative Proportion    0.6744   0.9650  1.00000 1.000e+00
```

```
pca$x
```

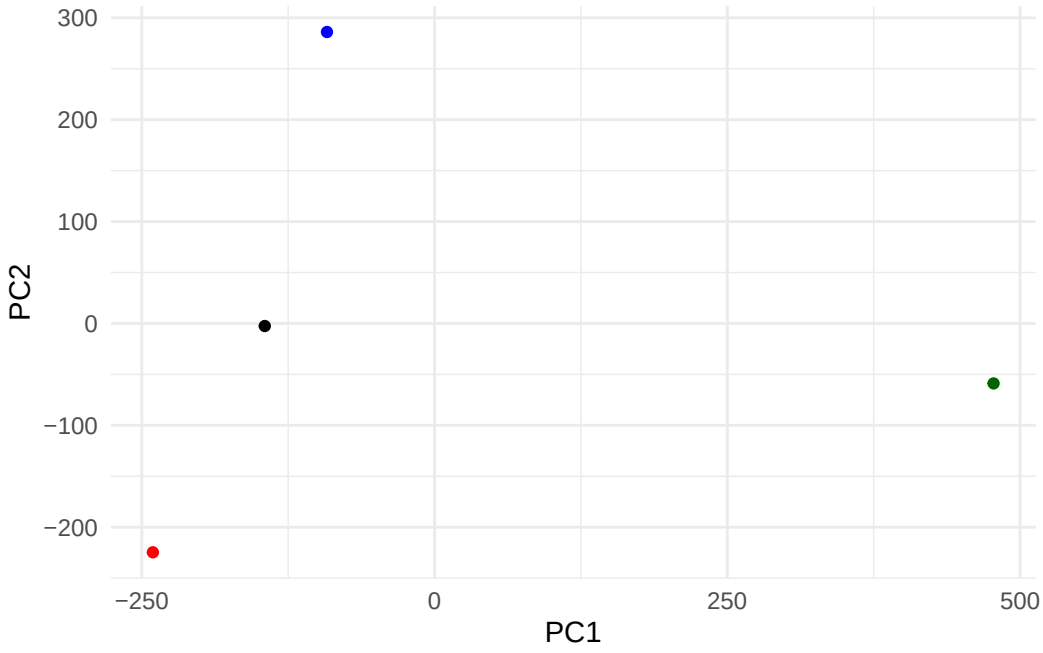
```
                PC1         PC2        PC3           PC4
England   -144.99315   -2.532999 105.768945 -9.152022e-15
Wales     -240.52915 -224.646925 -56.475555  5.560040e-13
Scotland   -91.86934  286.081786 -44.415495 -6.638419e-13
N.Ireland  477.39164  -58.901862  -4.877895  1.329771e-13
```

```r
cols <- c("black","red","blue","darkgreen")
plot(pca$x[,1], pca$x[,2], col=cols, pch=16, xlab="PCA1", ylab="PCA2")
```




```r
library(ggplot2)
```

```r
ggplot(pca$x) +
  aes(PC1, PC2) +
  geom_point(col=cols) +
  theme_minimal()
```

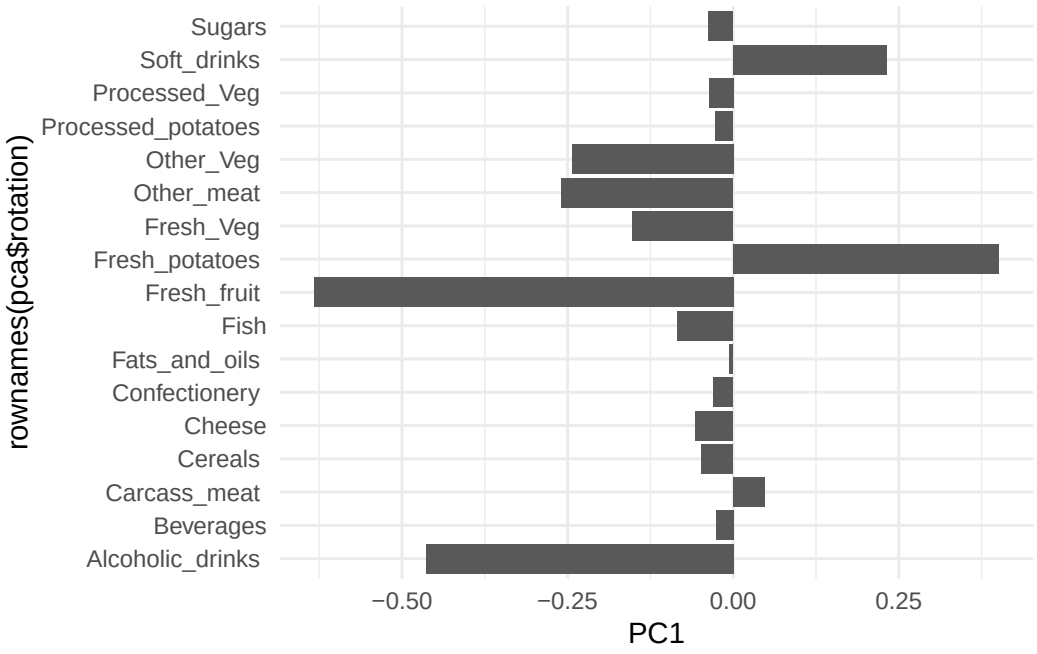


```
pca$rotation
```

```
                           PC1          PC2         PC3          PC4
Cheese              -0.056955380  0.016012850  0.02394295 -0.409382587
Carcass_meat         0.047927628  0.013915823  0.06367111  0.729481922
Other_meat          -0.258916658 -0.015331138 -0.55384854  0.331001134
Fish                -0.084414983 -0.050754947  0.03906481  0.022375878
Fats_and_oils       -0.005193623 -0.095388656 -0.12522257  0.034512161
Sugars              -0.037620983 -0.043021699 -0.03605745  0.024943337
Fresh_potatoes       0.401402060 -0.715017078 -0.20668248  0.021396007
Fresh_Veg           -0.151849942 -0.144900268  0.21382237  0.001606882
Other_Veg           -0.243593729 -0.225450923 -0.05332841  0.031153231
Processed_potatoes  -0.026886233  0.042850761 -0.07364902 -0.017379680
Processed_Veg       -0.036488269 -0.045451802  0.05289191  0.021250980
Fresh_fruit         -0.632640898 -0.177740743  0.40012865  0.227657348
Cereals             -0.047702858 -0.212599678 -0.35884921  0.100043319
Beverages           -0.026187756 -0.030560542 -0.04135860 -0.018382072
Soft_drinks          0.232244140  0.555124311 -0.16942648  0.222319484
Alcoholic_drinks    -0.463968168  0.113536523 -0.49858320 -0.273126013
Confectionery       -0.029650201  0.005949921 -0.05232164  0.001890737
```

```
# How much each of these variables weighs on each of the axes/pcs
# Telling us what makes two countries different from each other (based on PCA results)
```

```
ggplot(pca$rotation) +
  aes(PC1, rownames(pca$rotation)) +
  geom_col() +
  theme_minimal()
```


rownames(pca$rotation)
Sugars
Soft_drinks
Processed_Veg
Processed_potatoes
Other_Veg
Other_meat
Fresh_Veg
Fresh_potatoes
Fresh_fruit
Fish
Fats_and_oils
Confectionery
Cheese
Cereals
Carcass_meat
Beverages
Alcoholic_drinks
−0.50
−0.25
0.00
0.25
PC1